AI-based Research Companion (ARC): An Innovative Tool for Fostering Research Activities in Undergraduate Engineering Education

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***Abstract*—The engineering education today emphasizes the need to combine book learning with real-world application. However, much of the research done by undergraduates, which could be very valuable, is scattered and not fully used. To address this, a new tool called "AI-based Research Companion (ARC)" has been developed. ARC leverages advanced Generative AI technology, including GPT-4, to systematically organize, enhance, and offer personalized recommendations for undergraduate research projects. This platform is more than a simple tool; it aims to inspire undergraduates to dive into research by making the process approachable and engaging, thus increasing participation in research activities. Initial assessments of ARC have revealed an encouraging rise in student engagement with research, indicating a shift towards more research-oriented projects. The integration of GPT-4 within ARC stands out significantly; it precisely addresses the detailed demands of undergraduate research by providing a tailored, intelligent exploration pathway. By incorporating GPT-4's advanced features with a user-centric design, ARC emerges as an innovative platform, emphasizing the pivotal role of Generative AI in enhancing and expanding undergraduate research initiatives.**

***Index Terms*—AI-based Research Companion (ARC), Dynamic recommendations, Engineering education, Generative AI, GPT-4, Undergraduate research**

I. Introduction

The evolving landscape of Engineering education, rich in diversity, highlights the critical need for integrating classroom theory with real-world applications. This integration is often achieved through undergraduate research, which serves as a bridge connecting academic theories with their practical applications. However, in the digital age, a significant challenge has emerged: a vast amount of valuable undergraduate research remains fragmented and often overlooked.

In response to this challenge, we introduce the "AI-based Research Companion (ARC)", a platform that combines advanced technology with academic innovation to rejuvenate undergraduate research. ARC goes beyond being just a platform; it serves as a guiding light in the digital world, utilizing the power of Generative AI, especially models like GPT-4. ARC effectively navigates the vast ocean of research information, bringing together various aspects from detailed content analysis to user interaction. The result is a system that offers personalized, dynamic research recommendations closely aligned with the academic interests of students.

The importance of ARC is further highlighted in the context of AI's growing role in education. Recent studies, such as *Personalized Education in the Artificial Intelligence Era: What to Expect Next*, have shown the significant impact of AI in creating customized educational experiences [1]. This personalization trend is evident in specialized platforms like the *Design of AI-based self-learning platform for college English listening*, which focuses on enhancing specific skills such as English listening for college students [2].

Despite the progress in incorporating machine learning into e-learning platforms, as discussed in *A systematic review of machine learning techniques in online learning platforms* [3], the area of undergraduate research has not fully benefited from these technological advancements. Many studies, including *Artificial Intelligence in Education: A Review* [4], primarily focus on K-12 education or broader pedagogical approaches, leaving a gap in undergraduate research. ARC addresses this gap by integrating advanced Generative AI models like GPT-4 to provide finely-tuned, dynamic research recommendations.

ARC is designed to continuously learn and adapt, using the latest AI and machine learning technologies. This allows the system to not only grasp the extensive scope of engineering education but also to offer a personalized research experience for each student. By incorporating GPT-4, ARC achieves an unprecedented level of interaction and customization in educational tools, guiding students through a tailored academic research journey.

To enhance ARC's effectiveness, we conducted a compre-

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hensive data collection exercise at the university. We gathered detailed information about undergraduate students, including their research interests, current academic pursuits, challenges faced in research, and course specifics. This data formed a rich, comprehensive dataset that was fed into GPT-4, enabling it to understand and generalize the research preferences of different student profiles. Consequently, ARC can now suggest engaging research topics that align with students' interests, assist them at every step of their research journey, provide relevant literature, and support them in integrating their findings into their academic work. This holistic approach ensures that students are not only matched with suitable research topics but are also guided effectively throughout their research endeavors.

II. Literature Review

The advent of AI in education, particularly in the realms of personalized learning and e-learning systems, marks a transformative era in pedagogical methodologies. This literature review weaves through a tapestry of pioneering studies, each contributing a unique thread to the rich fabric of the AI-based Research Companion (ARC). These studies not only illuminate the path that led to ARC but also signal the dawn of an innovative future in educational technology.

The journey into personalized learning begins with Julianti et al. [6] and Qomariyah and Fajar [7], who underscore the necessity of tailoring educational experiences to individual learning styles. ARC emerges as an evolution of this concept, harnessing the power of AI to dynamically adapt to each student's academic inclinations.

As Rodrigues et al. [8] and Ahmad et al. [9] delve into the realms of educational data mining and AI applications in education, they highlight the expansive potential of data-driven insights. ARC taps into this potential, refining its algorithms to create a more intuitive and responsive learning companion.

In the works of Shishehchi et al. [10], Kolekar et al. [11], and Michel-Villarreal et al. [12], we see the versatility of AI in education taking shape through semantic recommendation systems, adaptive interfaces, and the nuanced applications of generative AI. These innovations lay the groundwork for ARC's sophisticated interface and its capability to navigate the complex landscape of academic research.

Bhutoria's [13] global perspective on AI in personalized education further expands the horizon, painting a picture of a world where education is not just a one-size-fits-all solution but a tailored journey for each learner. ARC is a manifestation of this vision, offering a tool that personalizes the academic research process to each student's unique needs.

Pividori and Greene's [14] exploration into AI-assisted academic authoring resonates with ARC's mission. Their work on streamlining the scholarly writing process mirrors ARC's goal of assisting students in their academic endeavors, ensuring clarity and quality in scholarly communications.

Finally, Glaese et al. [15] present Sparrow, an information-seeking dialogue agent. Their focus on creating dialogue agents that are helpful, correct, and ethically aligned is reflective of the principles guiding ARC. The development of Sparrow showcases the potential of dialogue systems in educational contexts, a potential that ARC seeks to harness.

In weaving together these diverse strands of research, ARC emerges not just as a product but as a harbinger of a new era in AI-driven education. It stands at the forefront of a movement towards more personalized, interactive, and intelligent educational tools. As one of the first to take this step, ARC paves the way for future innovations, signaling a future where AI not only complements but enhances the educational journey of learners worldwide.

III. Methodology and Features

This section outlines the comprehensive methodology employed in developing the AI-based Research Companion (ARC). It details the systematic processes involved in data collection and management, the integration of advanced AI technology, and the approaches taken to ensure an optimal user experience. Each subsection illustrates the deliberate steps taken to build ARC as a robust, AI-driven educational tool tailored to meet the dynamic needs of undergraduate engineering students. From gathering and processing significant datasets to embedding sophisticated AI models like GPT-4, and refining the user interface, this methodology forms the core of ARC's innovative design and functionality.

A. Data Collection and Management

The development of the AI-based Research Companion (ARC) was underpinned by a comprehensive data collection phase, focusing on two primary types of data that were crucial for training the AI model, specifically the Generative Pre-trained Transformer 4 (GPT-4):

- **Student Data:** Our approach involved engaging with undergraduate students across various engineering faculties. Through a series of meticulously designed surveys, structured interviews, and in-depth analysis of academic records, we collected a broad spectrum of data. This data encompassed:
  - Students' specific research interests and topics.
  - Details about their ongoing academic projects.
  - Challenges and hurdles faced during their research.
  - Comprehensive course-related information.

  This rich vein of information helped us understand the educational landscape from the students' perspective, offering insights into their preferences, obstacles, and academic pursuits.
- **Academic Research Materials:** Simultaneously, we compiled a vast array of academic research materials, including:
  - Scholarly articles and research papers.
  - In-depth case studies.
  - Detailed project reports.

  These materials spanned a wide range of engineering topics and were sourced from diverse platforms like academic journals, conference proceedings, and educational websites. The goal was to create a dataset that encapsulated the breadth and depth of engineering education.

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The combination of detailed student data with a diverse collection of academic research materials resulted in a multi-dimensional dataset, integral to the effective operation of the GPT-4 based ARC. This dataset was characterized by its:

- **Depth and Variety:** Encompassing a wide range of topics and student perspectives, the dataset provided a comprehensive understanding of the academic and research landscape in engineering education.
- **Structure and Organization:** The data was meticulously categorized and organized into a structured database. Personal data was anonymized to maintain confidentiality and adhere to ethical standards. This organization was crucial for efficient data retrieval and processing.

This structured and rich dataset was instrumental in training the GPT-4 model used in ARC. It enabled the AI to:

- Understand and analyze complex academic content and student queries.
- Identify and learn from patterns in student interactions and research preferences.
- Generate contextually relevant and personalized recommendations and responses.

Thus, the well-organized and comprehensive dataset was pivotal in empowering the GPT-4 based ARC to operate effectively, providing a tailored and dynamic educational tool that adapts to the evolving needs of undergraduate engineering students.

B. AI Integration and Recommendation System

The integration of AI into ARC, particularly the use of the Generative Pre-trained Transformer 4 (GPT-4) model, was a multifaceted and rewarding endeavor:

- **Training the AI Model:** The GPT-4 model was meticulously tailored using the collected dataset. This training process was crucial for the model to accurately grasp the intricacies of undergraduate research interests and the specific details of various engineering disciplines. By doing so, GPT-4 was enabled to discern patterns and nuances in both the academic content and student inquiries, allowing it to generate contextually relevant and academically appropriate recommendations.
- **Developing the Recommendation Algorithm:** The recommendation system of ARC was designed as a hybrid model, combining the strengths of collaborative filtering and content-based filtering. Collaborative filtering was employed to analyze and interpret user behavior and interaction patterns, while content-based filtering focused on the specific characteristics and relevance of the research materials. This dual approach was meticulously formulated to ensure the recommendations were not only relevant but also precisely tailored to align with each student's unique academic needs and interests.
- **Adaptive Learning and Iterative Improvement:** A defining feature of ARC is its capacity for adaptive learning. The system was built with the capability to continuously learn and adapt based on ongoing user interactions. This feature enables ARC to dynamically refine and personalize the learning experience for each user. Such adaptability ensures that the system's recommendations remain relevant, engaging, and effective over time, responding to the evolving academic landscape and student requirements.

C. User Experience

User experience was a primary focus in the development of ARC, aiming to create an interface that was both intuitive and effective. Inspired by conversational interfaces like ChatGPT, ARC's user interface was designed to be simple yet engaging. This approach was intended to provide a familiar and user-friendly experience, facilitating easy access to research materials and personalized recommendations.

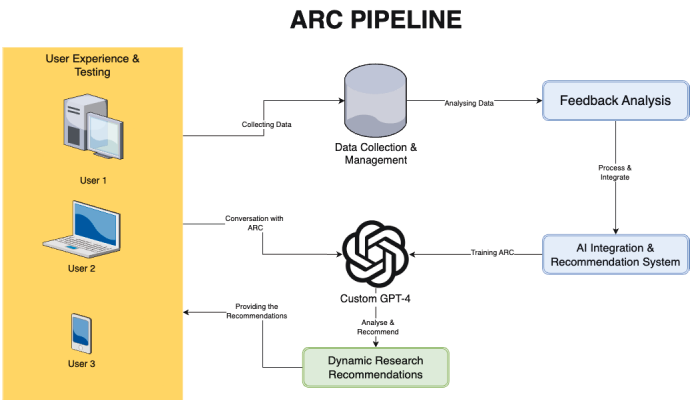

Fig. 1. ARC's Workflow

The ARC Workflow diagram (Figure 1) illustrates the comprehensive approach taken in developing the system. From the initial data collection phase to the ongoing adaptation based on user feedback, each stage plays a crucial role in ensuring the effectiveness of ARC as an educational tool. While still in its Minimum Viable Product (MVP) stage, this methodology lays a solid foundation for ARC, showcasing its potential to evolve and adapt to meet the dynamic needs of undergraduate engineering education.

D. Feature Development Based on Preliminary Feedback

As ARC embarked on its journey to revolutionize the research process for students in software engineering, computer science, and data science, it was crucial to align the platform's capabilities with the real needs of its users. Prior to the detailed "Initial Testing and Validation" phase, an iterative approach was adopted to refine ARC's features, emphasizing responsiveness to user feedback. Below are the key features that have been introduced:

- **Personalized Research Topic Suggestions:** Utilizing the GPT-4 model, ARC tailors research topic suggestions to align with users' academic interests, streamlining the research initiation process.
- **Interactive Q&A for Clarification and Learning:** This feature engages users in a dynamic dialogue, enabling

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them to explore complex topics and deepen their understanding effectively.

- **Manuscript Drafting and Revision Assistance:** ARC aids in the drafting and refinement of academic manuscripts, leveraging advanced natural language processing to enhance writing efficiency and quality.

Incorporating user feedback has been pivotal in ARC's evolution. A notable example of this user-driven development is the introduction of **Evidence-Based Recommendations**. This feature, currently in the planning stages, aims to enrich ARC's recommendations by underpinning them with cited evidence, enhancing both the credibility and relevance of the information provided to users. The commitment to incorporating such user-suggested enhancements reflects our dedication to continually evolve ARC, ensuring it meets the growing and changing needs of our academic community. Through this user-centric development process, ARC is poised to introduce more innovative features, further establishing itself as a leading AI-driven educational tool.

The discussed methodology and the features underpins ARC's effectiveness as an AI-driven tool in undergraduate engineering education. It combines detailed data collection, advanced AI integration, and a focus on user experience, ensuring that ARC not only meets but anticipates and adapts to the evolving needs of students and researchers.

IV. Initial Testing and Validation

During the Minimum Viable Product (MVP) stage, the AI-based Research Companion (ARC) underwent initial testing to evaluate its core functionalities. This critical phase engaged a controlled group of users from various engineering disciplines, enabling the collection of valuable data on ARC's performance and user satisfaction. The insights gained were pivotal for pinpointing areas for enhancement and affirming the AI recommendation system's efficacy.

A. Engineering Disciplines Tested

ARC was rigorously tested within specific domains of engineering education, particularly focusing on software engineering, computer science, data science, and related fields. The decision to concentrate testing efforts on these disciplines stems from their inherent research-intensive nature and the rapidly evolving landscape of technology they encompass. These areas often involve complex, multidisciplinary research challenges that are ideally suited for testing ARC's advanced AI-driven capabilities. By targeting these domains, we aimed to ensure that ARC's features were finely tuned and highly responsive to the unique demands of cutting-edge, technology-centric research. This focus allows for a thorough evaluation of ARC's ability to adapt and provide meaningful support in environments where the pace of innovation and the depth of inquiry present distinct challenges.

B. User Feedback and Observations

Feedback mechanisms played a crucial role in ARC's iterative development process. Users were encouraged to provide immediate feedback on their experiences, which was then meticulously analyzed to inform real-time system enhancements. This iterative feedback loop was essential for aligning ARC more closely with user needs.

- **Positive Feedback:** Users reported a marked increase in research efficiency and satisfaction, particularly highlighting ARC's intuitive design and tailored recommendations. The system's ability to facilitate easier access to relevant research materials and suggest topics of interest was frequently commended.
- **Areas for Improvement:** Despite the overall positive reception, users identified areas needing refinement. These included the need for greater accuracy in recommendation algorithms and the expansion of topic coverage to include more niche and technical areas within engineering research.

C. Beta Testing and Validation Participant Details

The beta testing phase engaged 50 undergraduate students, representing a broad cross-section of engineering specialties. These participants were selected through a combination of open calls and targeted recruitment, ensuring a comprehensive evaluation of ARC across different research contexts.

The outcomes from this initial testing phase were overwhelmingly positive, underscoring ARC's potential to significantly enhance undergraduate research experiences. Users particularly valued the platform's conversational interface and AI-driven suggestions, which collectively fostered a more engaging and productive research process. While the feedback highlighted specific areas for improvement, such as enhancing the recommendation system's precision and broadening the scope of covered topics, the overall consensus affirmed ARC's promise in advancing engineering education research. This validation not only endorses ARC's current capabilities but also charts a course for its future development and expansion.

V. Conclusion

To summarise, the introduction of GPT-4 in ARC represent a major change in the educational environment, particularly with respect to undergraduate research. Leveraging Generative AI, ARC aims to revolutionize and enhance undergraduate research accessibility. Early assessments, though limited, offer promising indications of ARC's potential to reshape student engagement in research, providing tailored and dynamic research pathways. As we move through the challenges of the digital age, platforms such as ARC, driven by detailed methods and a focus on the user, are set to lead us towards a future filled with extensive and connected research projects. Continuous enhancements and more extensive testing will strengthen ARC's contribution to undergraduate engineering education, promoting a lively environment of scholarly exploration and innovation.

References

[1] M. Murtaza, Y. Ahmed, J. A. Shamsi, F. Sherwani and M. Usman, "AI-Based Personalized E-Learning Systems: Issues, Challenges, and Solutions," in IEEE Access, vol. 10, pp. 81323-81342, 2022, doi: 10.1109/ACCESS.2022.3193938.

Authorized licensed use limited to: REVA UNIVERSITY. Downloaded on October 13,2025 at 10:55:17 UTC from IEEE Xplore. Restrictions apply.

[2] J. Zhou, ”Design of AI-based self-learning platform for college English listening,” 2020 2nd International Conference on Machine Learning, Big Data and Business Intelligence (MLBDBI), Taiyuan, China, 2020, pp. 544-547, doi: 10.1109/MLBDBI51377.2020.00114.

[3] C. E. K. Agbewali-Koku, M. A. Rahman, M. Hamada, M. A. Ali, L. N. Oysharja and M. T. Hossain, ”A systematic review of machine learning techniques in online learning platforms,” 2022 IEEE 15th International Symposium on Embedded Multicore/Many-core Systems-on-Chip (MCSoC), Penang, Malaysia, 2022, pp. 247-250, doi: 10.1109/MCSoC57363.2022.00046.

[4] L. Chen, P. Chen and Z. Lin, ”Artificial Intelligence in Education: A Review,” in IEEE Access, vol. 8, pp. 75264-75278, 2020, doi: 10.1109/ACCESS.2020.2988510.

[5] W. G. C. S, S. S. G. T. S, B. K. H. P and A. Gamage, ”AI and Machine Learning Based E – Learning System For Secondary Education,” 2022 IEEE 7th International conference for Convergence in Technology (I2CT), Mumbai, India, 2022, pp. 1-6, doi: 10.1109/I2CT54291.2022.9824643

[6] M. R. Julianti, Y. Heryadi, B. Yulianto and W. Budiharto, “Recommendation System Model for Personalized Learning in Higher Education using Content-Based Filtering Method,” 2022 International Conference on Information Management and Technology (ICIMTech), Semarang, Indonesia, 2022, pp. 1-6, doi: 10.1109/ICIMTech55957.2022.9915109.

[7] N. N. Qomariyah and A. N. Fajar, “Recommender System for e-Learning based on Personal Learning Style,” 2019 International Seminar on Research of Information Technology and Intelligent Systems (ISRITI), Yogyakarta, Indonesia, 2019, pp. 563-567, doi: 10.1109/ISRITI48646.2019.9034568.

[8] M. Wander Rodrigues, S. Isotani, L. E. Zárate, “Educational Data Mining: A review of evaluation process in the e-learning,” Telematics and Informatics, Volume 35, Issue 6, 2018, Pages 1701-1717, ISSN 0736-5853, doi: 10.1016/j.tele.2018.04.015.

[9] K. Ahmad et al., “Data-Driven Artificial Intelligence in Education: A Comprehensive Review,” in IEEE Transactions on Learning Technologies, doi: 10.1109/TLT.2023.3314610.

[10] S. Shishehchi, S. Y. Banihashem and N. A. M. Zin, “A proposed semantic recommendation system for e-learning: A rule and ontology based e-learning recommendation system,” 2010 International Symposium on Information Technology, Kuala Lumpur, Malaysia, 2010, pp. 1-5, doi: 10.1109/ITSIM.2010.5561329.

[11] S. V. Kolekar, S. G. Sanjeevi and D. S. Bormane, “Learning style recognition using Artificial Neural Network for adaptive user interface in e-learning,” 2010 IEEE International Conference on Computational Intelligence and Computing Research, Coimbatore, India, 2010, pp. 1-5, doi: 10.1109/ICCIC.2010.5705768.

[12] Michel-Villarreal, R.; Vilalta-Perdomo, E.; Salinas-Navarro, D.E.; Thierry-Aguilera, R.; Gerardou, F.S. “Challenges and Opportunities of Generative AI for Higher Education as Explained by ChatGPT,” Educ. Sci. 2023, 13, 856. doi: 10.3390/educsci13090856.

[13] Aditi Bhutoria, Personalized education and Artificial Intelligence in the United States, China, and India: A systematic review using a Human-In-The-Loop model, Computers and Education: Artificial Intelligence, Volume 3, 2022, 100068, ISSN 2666-920X, doi: 10.1016/j.caeai.2022.100068

[14] Pividori, M., & Greene, C. S. (2023). A publishing infrastructure for AI-assisted academic authoring. In bioRxiv. https://doi.org/10.1101/2023.01.21.525030

[15] Glaese A, McAleese N, Trebacz M, Aslanides J, Firoiu V, Ewalds T, Rauh M, Weidinger L, ... 18 more ..., Young S, Gabriel I, Isaac W, Mellor J, Hassabis D, Kavukcuoglu K, Hendricks LA, Irving G. Improving alignment of dialogue agents via targeted human judgements, 2022. In arxiv: https://doi.org/10.48550/arXiv.2209.14375

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